



Partial Exam Answers EN December 2021

Macroeconomia / Macroeconomics (Università Commerciale Luigi Bocconi)

Your grade will be notified on
Wednesday, January 12, 2022.

Question 1 (7.5 points)

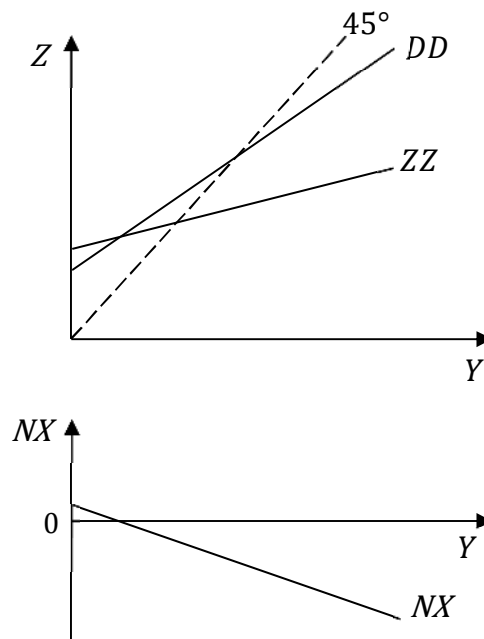
In country XYZ there are only one-year and two-year bonds. Current and expected future inflation are equal to zero, so nominal and real interest rates are the same. In period $t + 1$ ("the future"), consumption depends only on disposable income at time $t + 1$, while investment depends on the values that interest rate and income will assume at time $t + 1$. As for the current period (t , "the present"), consumption depends positively on disposable income at time t and on the expected disposable income at time $t+1$, $C_t = C(Y_t - \bar{T}_t, Y_{t+1}^e - \bar{T}_{t+1}^e)$, investment depends positively on current and expected future income levels, and negatively on the current and future expected levels of short-term interest rates, the central bank chooses the interest rate, and G and T are exogenous as usual. Furthermore, at time t , in the initial equilibrium position both income and the interest rate take on positive values, while at time $t + 1$ the economy is expected to be in a 'liquidity trap'. Suppose now that, at time t , the government announces that at time $t + 1$ taxes will be increased. At the same time, the central bank announces that it will adjust the policy rate so as to prevent any change in the equilibrium level of income for time t , and time t only (Y_t). Explain how these fiscal and monetary policy announcements, which come as a surprise to individuals and are believed, will change the yield to maturity of the two-year bonds at time t , i_{2t} .

SKETCH OF ANSWER

The tax increase at time $t + 1$ will shift the IS for that period to the left, leading to an equilibrium characterized by a lower level of income (Y_{t+1}) and an interest rate ($i_{1,t+1}$) still equal to zero (in fact, being the economy in a liquidity trap at time $t + 1$, the intersection between the new IS and the LM curve will continue to take place for an interest rate equal to zero). The reduction of expected future income and the expected future tax increase will reduce consumption demand at time t , thus causing a leftward shift of the IS. To prevent Y_t from falling, the central bank will have to cut the time t policy rate. It follows that, at time t , the economy will reach an equilibrium in which Y_t is unchanged and $i_{1,t}$ is lower. Since i_{2t} is (approximately) equal to the average of the short-term current and expected future interest rates, and that $i_{1,t+1}$ remains equal to zero while $i_{1,t}$ falls, the yield to maturity of the two-year bonds at time t will certainly decrease.

Question 2 (7.5 points)

Felicia is an open economy where no financial markets exist, the Marshall-Lerner condition is met, and whose initial equilibrium is depicted in the figure below. Explain if, in this economy, the level of income for which the goods market is in equilibrium is greater, smaller, or equal to the level of income for which net exports are equal zero. Furthermore, suggest a policy mix that would allow Felicia to reach a new equilibrium with an unchanged equilibrium income and a net exports surplus. When answering, make sure to describe verbally how the policy mix you propose affects the position in the plane of each of the three curves represented in the graph.



SKETCH OF ANSWER

The figure allows one to conclude that, in this economy, equilibrium income (the value of Y for which the ZZ curve cuts the 45° ray emanating from the origin) is greater than the level of income for which net exports are balanced (the value of Y for which the ZZ and the DD intersect each other – or, equivalently, the value of Y for which the NX curve crosses the x -axis). In order to transform into a surplus the initial net exports deficit while keeping income unchanged, one has to implement a combination of fiscal contraction and exchange rate depreciation. Due to the fiscal tightening, the DD and the ZZ curves will shift downwards by the same amount. On the other hand, and given that the Marshall-Lerner condition is met, the exchange rate depreciation will raise net exports, causing both the NX and the ZZ curves to shift upwards. It follows that, through an appropriate combination of the two interventions – in particular, one that leaves in the original position the ZZ curve and leading to a DD curve that intersects this unchanged ZZ below the 45° line – both goals can be achieved. [Intuition behind this result: the fiscal contraction lowers equilibrium income, but improves the trade balance; the exchange rate depreciation raises both net exports and equilibrium income. Through an appropriate combination of fiscal tightening and exchange rate depreciation, one can reach an equilibrium where income is unchanged, while net exports have become positive].

Question 3 (7.5 points)

Consider an open economy under a flexible exchange rate regime. Domestic and foreign prices are constant and for simplicity both equal to 1 ($P = P^* = 1$), consumption depends positively on disposable income, investment is entirely exogenous ($I = \bar{I}$), net exports depend positively on foreign income, negatively on domestic income, and positively on the real exchange rate, so that the Marshall-Lerner condition is not met. Finally, G and T are exogenous, and the country's central bank chooses the domestic interest rate. Having discussed if, and why, the IS curve of the "IS-LM-interest parity condition" model that describes this economy will have a negative, zero or positive slope, suppose that, starting from an initial equilibrium position, E^e falls by $\Delta E^e < 0$ and that, to prevent this change in the future expected exchange rate from affecting the equilibrium level of income, the government intervenes by changing the level at which net taxes are set. To make sure that equilibrium income will remain constant, should the government increase or decrease net taxes? Following these changes in E^e and net taxes, which of the three curves (the IS, the LM, and that representing the interest parity condition) will shift in the plane where they are drawn, and in which direction? Finally, discuss how the exchange rate, money supply, money demand, consumption, investment, government purchases of goods and services, and net exports will have changed in the new equilibrium that the economy will reach. Explain.

SKETCH OF ANSWER

The IS will have a positive slope. In fact, if – starting from a goods market equilibrium position – the interest rate goes up, investment does not change (it is exogenous), the exchange rate will appreciate (via interest parity condition) and net exports will increase, since they depend positively on the real exchange. To go back to goods market equilibrium, the supply of goods will have to rise. It follows that goods market equilibria with higher levels of i are in this case associated with higher levels of Y – the IS curve will be positively sloped in the (Y, i) plane.

As for the effects of the fall in E^e , it will shift up and to the left the curve representing the interest parity condition. For given domestic and foreign interest rates, the exchange rate consistent with this latter condition will now be lower. This exchange rate depreciation will decrease net exports (recall that the Marshall-Lerner condition is not met), thus shifting the IS to the left. To prevent income from falling, the government will have to cut net taxes by the amount needed to bring the IS curve back to its original position. Since Y and i do not change, money demand will remain unchanged, and the same is true about money supply. Finally, consumption will go up (due to the tax cut) and exports down (because of the depreciation of the exchange rate, since the Marshall-Lerner condition does not hold), while I and G (both entirely exogenous) will be unchanged.

Question 4 (7.5 points)

As you know, the debt-to-GDP ratio ($b \equiv B/Y$) evolves over time as implied by the following equation:

$$b_t - b_{t-1} = (r - g)b_{t-1} + d \quad (*)$$

Describe the parameters entering the above equation and explain what is meant by steady state debt-to-GDP ratio ($\bar{b}$), providing the analytical expression for this particular value of b . Finally, use equation (*) to explain if and why you agree, or do not agree, with the following statement:

“In a country that, at time t , inherits from the previous period a debt-to-GDP ratio $b_{t-1}^0 > 0$ and in which $r < g$ and $d = 0$, b will always converge to its steady state value, and this even for very high values of b_{t-1}^0 ”.

[N.B.: you are not been asked to draw the graph used to analyze the time evolution of the debt-to-GDP ratio, but rather to answer by making reference to equation (*) only. Answer that will not make explicit reference to that equation will receive zero points.]

SKETCH OF ANSWER

The steady state debt-to-GDP ratio $\bar{b}$ is the value at which, once attained, b will remain constant overtime. It is equal to the ratio between d (the primary deficit-to-GDP ratio) and the difference between the growth rate of the economy (g) and the real interest rate (r) [as it is straightforward to verify by setting $b_t - b_{t-1} = \bar{b}$ in (*), and then solving for $\bar{b}$]. To conclude that the statement is correct, just note that, when $d = 0$, then $\bar{b} = 0$ and (*) becomes $b_t - b_{t-1} = (r - g)b_{t-1}$. Being $r < g$, for any $b_{t-1} > 0$ inherited from the previous period the right-hand side of this latter equation is negative, which in turn implies $b_t < b_{t-1}$ for all t . It follows that the debt-to-GDP ratio will keep falling period after period, converging to its steady state value (zero). The reason is that, when the primary deficit is zero, the stock of government debt grows over time at the rate r , while income, the variable that appears at the denominator of the debt-to-GDP ratio, grows at the rate g . When $r < g$, b is therefore bound to fall over time and to converge to zero, and this independently of the value of the debt-to-GDP ratio inherited from the past.

